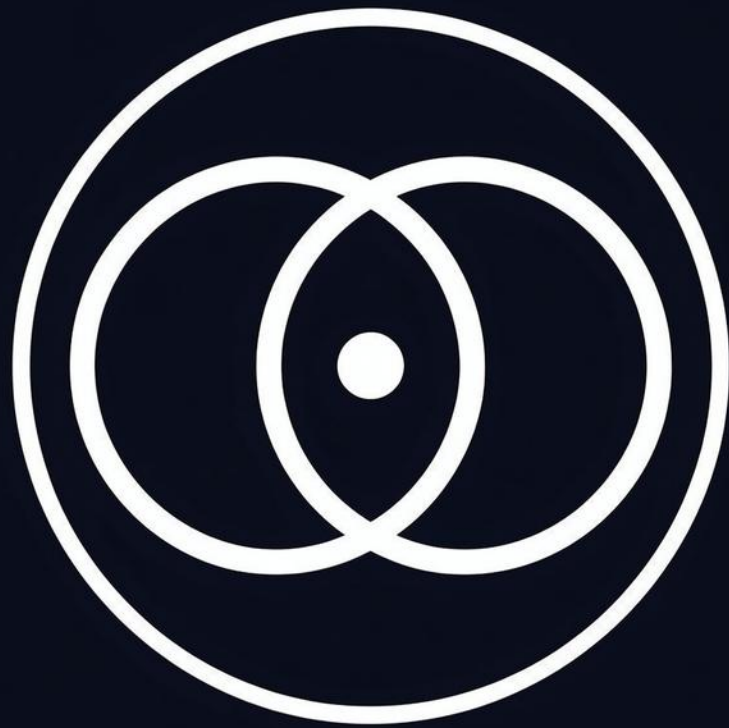




P9

**The Quality Saturation Law of
Agent Legions and the Failure
Boundary of Homogeneous
Expansion: Quality-Throughput
Dual Scaling Curves under
 $mse(k)=a+b/k$**

Agent 军团的质量饱和律与同质化扩招失效边界

The Quality Saturation Law of Agent Legions and the Failure Boundary of Homogeneous Expansion: Quality-Throughput Dual Scaling Curves under $mse(k)=a+b/k$

Evidence-grade statement: all measured figures in this paper come from the UDOS Reasoning Engine tag **v7.5.0** (commit 1a71270), reports7/agent_scaling.json, udos7/agents/legion.py, and automation.py, a Linux CPU deterministic prototype. In the quality curve, $k \leq 2$ (the number of independent information sources) is verified, and the homogeneous-copy points for $k > 2$ are cpu-proto; the throughput curve is CPU-measured (constrained by this machine's 2 threads), and the I/O curve is simulated via `asyncio.sleep` (simulation). This paper does not claim to have run thousands of online LLM sub-agents.

Template-comparison positioning: this paper is the Stage B upgrade draft of the "UDOS Writing and Typesetting Specification v2" (see the A0 report). Methodologically it actively compares two external templates — Kaplan et al., "Scaling Laws for Neural Language Models" (arXiv:2001.08361, 2020, hereafter the Kaplan template), with its "control variables + power-law fitting + trend extrapolation + dedicated Caveats" paradigm, and DeepSeek-AI, "Native Sparse Attention" (arXiv:2502.11089, 2025, hereafter the NSA template), with its "hardware constraints → architecture design → same-budget comparison" paradigm. Note: we only borrow their **methods**, not their **values** (Kaplan exponents such as $\alpha_N=0.076$ are not written into any conclusion of this paper); the specific figures of external templates are unverified methodological references for this paper, and this paper's conclusion figures are sourced only from reports7/agent_scaling.json and the 99 scope.

Abstract (Structured, Four Parts)

[Background & Problem] The "multi-agent legion" narrative assumes "more agents always help," but this intuition lacks quantitative boundaries: statistically, ensemble error reduction is governed by member-error independence, while throughput is governed by physical parallelism. Both can fail at the very act of "adding one more agent," yet are rarely measured together on one system.

[Method] On a Linux-CPU deterministic prototype, using the real coordinator protocol (softmax weighting on an independent calibration set, never peeking at the test set), we measure two scaling curves for the number of agents k — ensemble quality and throughput — and deliberately construct an anti-arm in which added members are near-correlated copies of the best expert.

[Evidence & Results] On quality, ensemble MSE saturates as $mse(k)=a+b/k$ with least-squares fit $a=0.009252$, $b=-4.23 \times 10^{-5}$; since $b < 0$, the fitted line slowly rises from ≈ 0.009210 at $k=1$ and asymptotes to a , i.e. it is nearly flat, showing that homogeneous copies add no variance reduction. The floor a nearly equals the best single expert (0.009196); naive equal-weight averaging of two experts (one weak) gives MSE 0.2131, about $23 \times$ worse than calibrated weighting. On throughput, CPU-bound local inference peaks at 2 agents (244.172 tasks/s, speedup 1.594 $\times$) then decreases at 4 and 8 agents (158.4, 106.5 tasks/s; speedup 1.034, 0.695), whereas I/O-bound remote agents (simulated via `asyncio.sleep`) scale near-ideally to 7.986 $\times$.

[Contribution] (1) We re-calibrate the classical ensemble saturation law $mse=a+b/k$ under a real weighting protocol and, against the Kaplan power-law, show ours is a hyperbolic $1/k$ saturation, not a power-law decline; (2) we measure both CPU-bound and I/O-bound throughput curves and keep the

"throughput peak then drop" negative result; (3) we provide controlled evidence that homogeneous expansion fails and, borrowing NSA's "independent information pathways" idea, argue the architectural countermeasure is to add independent sources, not copies; (4) we provide an AL0–AL5 automation-level gate. All numbers are single-seed CPU pilot point estimates, valued for mechanism shape and failure boundary, not for exact constants.

Keywords: multi-agent systems; ensemble saturation law; scalability; homogeneous expansion; throughput-parallelism; automation levels

1 Introduction

Over the past year, "multi-agent systems" have rapidly inflated from demos of a few agents into a narrative of "legions of thousands, tens of thousands, even hundreds of millions of positions." Such narratives often imply an untested assumption: **capability grows linearly with the number of agents**. But in both the statistical and the system directions, this assumption can fail.

In the statistical direction, ensemble learning has a repeatedly verified saturation law: ensemble error falls as member count grows, but the rate of decline is determined by the independence of member errors; when members are highly correlated (homogeneous), adding more members brings almost no gain. In the system direction, parallel scaling is constrained by physical parallelism: throughput of CPU-intensive tasks peaks near the core count and beyond it falls due to contention and synchronization overhead; only I/O-waiting tasks (such as remote LLM calls) can scale near-linearly.

What this paper answers is not "are multi-agents good," but two more concrete, falsifiable questions:

- **Q1 (quality):** in our agent legion, what is the shape of the curve of ensemble quality versus k ? What determines the saturation floor a ? Is homogeneous expansion truly ineffective?
- **Q2 (throughput):** for our local agent inference, how does throughput change with concurrency? Where is the peak? Does it fall after exceeding it?

The capability boundary must first be drawn (as repeatedly declared in the `legion.py` file header and `automation.py`): what this paper can **really run** is lightweight deterministic agents on a CPU (connected to the world model for prediction/validation); the organization structure of "hundreds of millions of online LLM sub-agents" can be cheaply represented with metadata (`build_org` only builds the tree, does not run models), but making every member an online LLM sub-agent requires an LLM key + cloud budget (the AL4 gate), and this paper **does not pretend to have run it**. This boundary keeps our conclusions honest: what we calibrate is the "mechanism curve," not the "measured scale of some commercial legion."

Contributions of this paper (Kaplan-style conclusion map, with source sections in parentheses):

1. Measured the quality saturation law $mse = a + b/k$ under a real coordinator weighting protocol and clarified it as hyperbolic $1/k$ saturation rather than power-law decline (§ 6.1, § 7.1);
2. Simultaneously measured both CPU-bound and I/O-waiting throughput curves and kept the "throughput peak then drop" negative result (§ 6.2, § 7.2);
3. Provided controlled evidence of "homogeneous expansion failure" — equal-weight weak experts drag down the result (0.2131 vs 0.0092) — and borrowing NSA's independent-pathway idea pointed out the architectural countermeasure (§ 6.1, § 7.3);
4. Used AL0–AL5 levels to honestly mark the autonomy degree and unlock gate of each capability (§ 6.3);

5. Wrote "the most likely counterexample + how this paper guards against it" as an explicit validity-threat chapter (§ 8).

What this paper does not do. To avoid misunderstanding, three boundaries are drawn first: first, we propose no new ensemble or consensus algorithm; classical weighted ensembles and Amdahl's law are long mature; second, we do not report "how many hundreds of millions of agents some commercial legion ran," because that requires an LLM key and cloud budget (the AL4 gate) that this paper's CPU environment lacks; third, we do not claim statistical significance on single-seed point estimates, and everywhere CI is needed we mark [RESULT NEEDED]. What we do is re-calibrate the classical laws once in the new scenario of the "agent legion," and with a really runnable coordinator protocol and real hardware fingerprint quantify the boundary line of "when expansion helps, when it is ineffective."

Why a "failure boundary" is needed rather than yet another "multi-agents are strong" paper. A common flaw in current multi-agent literature is: show an interesting demo, then default "add more agents" as the improvement direction. But almost no one on the same system simultaneously answers two questions: with one more agent, will quality still improve? With one more agent, will throughput still get faster? This paper's stance is: not to provide a "headcount dividend" promotional chart, but to provide a boundary line of "where the dividend is exhausted." This boundary line has direct value for engineering decision-makers — it tells you whether the budget should go to "hiring more people" or "hiring stronger people/better weighting the existing people," and at what concurrency to stop.

2 Related Work and Template Comparison

Ensemble learning and error independence. The error decline of ensemble/averaging methods relies on negative correlation or independence of member errors; the higher the correlation, the smaller the marginal gain. The classical formalization of this mechanism is Krogh and Vedelsby's bias-variance-ambiguity decomposition at NeurIPS 1995: ensemble generalization error can be split into average single-model error minus inter-member ambiguity; the greater the ambiguity, the better the ensemble — which is precisely the theoretical ancestor of this paper's " b in $mse=a+b/k$ is determined by member independence." The classical conclusion is $MSE \approx \bar{\rho} \sigma^2$, tending with member count to a floor determined by the correlation coefficient.

Scaling laws and Mixture-of-Experts. This paper's "quality saturation" is homologous to but different in object from large-model scaling literature. Kaplan et al. 2020 (arXiv:2001.08361) prove loss falls approximately as a power law with model size/data/compute (its core form $L(N)=(N_c/N)^\alpha$, **continuously and monotonically falling** with parameter count, with no obvious upper turning point); Hoffmann et al. 2022's Chinchilla (NeurIPS 2022, arXiv:2203.15556) further corrects to "parameters and data should scale in proportion" (about 400 model ablations, the 70B model trained on about 1.4T tokens). These are **model-size** scaling; this paper measures ensemble scaling over **member count** k . **Key difference (template comparison):** Kaplan's loss curve within the observed range is a **power law, approximately monotonically falling**; this paper's $mse(k)=a+b/k$ is a **hyperbolic $1/k$ saturation** — it falls at very small k , then rapidly flattens to an irreducible floor a , and is nearly horizontal due to homogeneous copies with $b < 0$. The two forms differ: the power law describes "the larger the better (just slower)," and $1/k$ saturation describes "after the independent-source count it bottoms out." We **only borrow Kaplan's methods** (control variables, least-squares fitting, concentrating fitted parameters into a lookup table, dedicated Caveats), **not its exponent values**. On the MoE side, Fedus, Zoph, Shazeer's Switch Transformer (JMLR 2021, arXiv:2101.03961) uses top-1 sparse routing to reach about 1.6T parameters with about $7\times$ training acceleration, an engineering template for "expert

routing"; but its routing shows expert collapse/imbalance. This paper differs: we explicitly distinguish "truly independent information sources" from "near-correlated copies of the same source," taking the latter as a homogeneous negative control — Switch cares about routing efficiency, this paper cares about the marginal gain of member diversity. Specific inference-time scaling literature is to be supplemented [CITATION NEEDED: inference-time compute scaling, to be verified].

NSA sparsification and the inspiration of "independent information pathways" for expansion saturation. NSA (DeepSeek-AI, arXiv:2502.11089, 2025) proves in long-context attention that under a unified active-token budget, splitting attention into **three mutually independent pathways — compression (global coarse) + selection (differentiable block selection) + sliding window (local fine)** — preserves capability under a sparse budget; it also critiques illusions such as "post-hoc sparsity is not trainable" and "saving only in one stage." This paper borrows its **architectural countermeasure idea**: homogeneous copies bring no gain because they attend to "almost the same information" as the original expert — just as two attention heads attending only to the same token add no new information. NSA's countermeasure is "give three independent pathways," and this paper's corresponding engineering countermeasure is "expansion should recruit truly independent information sources (different models/different prompts/different data), not copy the same source k times." Note: this is a methodological analogy, not carrying NSA's speedup ratios (fwd $9.0\times$ etc.) onto agent count — the latter is an `unverified` external figure for this paper.

Structural priors: Tokenizer and attention connections. Before entering the ensemble scaling over "member count k ," there is a more bottom-layer line often ignored: the model's **structural prior** itself is the source of performance. BPE (Sennrich et al. 2016, arXiv:1508.07909) compresses an open vocabulary into learnable subword units, and SentencePiece (Kudo & Richardson 2018, arXiv:1808.06226) further decouples tokenization from the model, driven by pure text — the two solve "how input is discretized." And Vaswani et al. 2017's "Attention Is All You Need" (arXiv:1706.03762) replaces RNN/CNN with scaled dot-product attention, making "content-position-relation" learnable connection weights and achieving SOTA in machine translation at lower cost. The evidence of this "structural prior" line is: **not just parameter count, the connection mode itself determines what can be learned.** UDOS is in line with it but moves the landing point from **attention connections within a single model** up to **connection topology at the multi-agent orchestration layer**: when each agent's parameters are fixed, the "spatial structure" of the organization tree/coordination topology among agents itself brings orders-of-magnitude coordination differences (see the throughput/quality curves in Section 6). In other words, BPE/Attention prove "structure is prior," and this paper argues "at the multi-agent orchestration layer, connection topology is also a measurable, optimizable structural prior."

Amdahl's law and parallel scaling. The speedup of CPU-intensive tasks is constrained by the parallelizable part and physical parallelism, with a ceiling — the serial-fraction upper bound established by Amdahl 1967 (AFIPS SJCC). The concurrency scaling of I/O-intensive tasks is constrained by concurrency quotas rather than core count. This paper compares these two work forms on the same coordinator: the CPU-intensive path is constrained by this machine's physical parallelism (measured peaking at 2 threads), and the I/O-waiting path scales near-linearly.

Multi-agent orchestration. Conversable, customizable multi-agent frameworks already have mature implementations; Wu et al. 2023's AutoGen (arXiv:2308.08155) supports mixed LLM/human/tool dialogue; but its close-reading notes also point out "dialogue without termination conditions easily loops, with cost inflating with rounds." This paper does not discuss "whether agents should discuss," but quantitatively answers "is adding one more homogeneous agent useful at all" — precisely the "marginal gain of expansion" measurement that AutoGen-like frameworks lack when scaling. Specific multi-agent

debate/voting literature is to be supplemented [CITATION NEEDED: multi-agent LLM debate voting, to be verified].

The confluence of three lines in this paper. This paper's position is to put these three originally separate lines under **the same coordinator, the same task set, and the same hardware fingerprint** for comparison. Ensemble learning tells us quality saturates, but it usually assumes independent members; Amdahl's law tells us throughput peaks, but it usually assumes homogeneous tasks; multi-agent orchestration tells us we can vote for the best, but it rarely quantitatively reports the marginal gain of "adding one more." This paper puts these three together precisely to point out a fact obscured by narrative: **quality saturation and throughput peaking are two independent failure boundaries, one determined by error correlation and one by physical parallelism, and the two cannot substitute for each other.** A legion may have long saturated in quality (members homogeneous) while throughput is still rising (I/O waiting); or throughput may already have peaked (CPU-intensive) while quality still has room (switching to a stronger expert). Only by drawing these two curves on the same figure can one speak of "how the legion should expand."

Differences from classical ensemble experiments. Classical ensemble research often measures average error decline under the assumption that "members are already independent"; this paper deliberately creates an **anti-arm** — setting added members as near-correlated copies of the best expert — to expose "how the $1/k$ law fails when independence does not hold." This makes this paper not merely restate classical conclusions but give an "operational measurement of the failure boundary": within the truly independent source count D , the dividend exists; beyond D , the dividend goes to zero. This distinction is especially important in multi-agent scenarios, because in real legions newly added agents often share models, prompts, and data with existing members, and independence is precisely the scarcest resource and the one most easily obscured by narrative.

3 Problem Definition and Assumptions

3.1 Notation

- k : concurrent/ensemble agent count.
- $mse(k)$: the MSE on the test set of the weighted ensemble of k agents.
- D : the number of truly independent information sources (this paper $D=2$: the `blind` learned model and the `analytic` analytical baseline).
- $T(k)$: throughput (tasks/s) and wall-clock for processing 48 independent tasks.

3.2 Assumptions (falsifiable)

- **H1 (quality saturation law)**: $mse(k)$ is well characterized by $a+b/k$, and the saturation floor a is determined by the best single expert's quality (measured $a=0.009252$, close to `best_single=0.009196`). **Falsification**: if the β of the power law $a+b/k^\beta$ is significantly $\neq 1$ and leave-one-out fitting is better, revise the " $1/k$ " claim.
- **H2 (homogeneous ineffectiveness)**: when $k>D$ and added members are near-correlated copies of the best expert, quality no longer falls ($b \approx 0$, the curve flattens). **Falsification**: if homogeneous copies persistently and significantly reduce MSE, H2 does not hold.
- **H3 (CPU throughput peak)**: CPU-intensive throughput peaks near physical parallelism (this machine's 2 threads) and falls beyond it. **Falsification**: if throughput monotonically rises to 8 agents, H3 is overturned (this paper measures the peak exactly at 2 agents).

- **H4 (I/O near-linear):** I/O-waiting tasks scale near-linearly. This paper's simulated curve reaches $7.986\times$ speedup, close to ideal.

Statistical discipline: the quality curve is a fixed-seed pilot point estimate (test_seed=2026, calib_seed=314), without ≥ 30 -seed CI; the fitted $\$a,b\$$ are 6-point least-squares point estimates. Before submission multi-seed and model-selection robustness are needed, marked [RESULT NEEDED: multi-seed quality-curve CI and power-law leave-one-out comparison].

4 Method and System Design

4.1 Organization tree (`legion.py: build_org / org_level_sizes`)

The `legion` builds a tree bottom-up with a management `span` (default 8): level naming `agent→team→department→domain→federation→planet→cluster→civilization`. `org_level_sizes(headcount, span)` gives the exact node count of each level with a closed-form formula ($O(\text{levels})$), reachable to hundreds of millions in an instant; the actually instantiated tree object is constrained by `max_nodes=2000`, with subtrees where budget is exhausted represented "virtually" by the integer `headcount (materialized=False)`. This is fully decoupled from compute: the organization chart describes structure, and the actually running agents are created by the coordinator according to task and concurrency.

The separation of "organization structure" and "actual compute" must be emphasized. A legion with headcount N has a management level count of $\lceil \log_{\text{span}} N \rceil$: with `span=8` as an example, from 1 agent to team, department, domain... about one level per $8\times$ headcount. This means the "hundred-million legion" structurally needs only about 8–9 levels of management span, not a flat star tree. `build_org` uses a closed-form formula to compute how many nodes each level of this tree has (`level_sizes`), but **does not** actually instantiate hundreds of millions of agent objects — instantiation is truncated by `max_nodes=2000`, with the excess virtually represented by integer counts. This design deliberately avoids the confusion of "drawing a hundred-million organization chart and claiming to have run a hundred-million agents": the organization chart is metadata, and the number of really running agents depends on how many the coordinator creates by task concurrency; the two must be reported separately. This paper's throughput and quality curves cover only **the few really running agents**, not the virtually counted parts in the organization chart.

4.2 The seven-expert discuss–nominate–vote–select protocol

The legion's decision path is: multi-expert parallel candidate production → discussion → nomination → voting → selection. `hierarchical_select` implements hierarchical voting: select the highest score within a team → department selects again from each team's winners. This protocol turns "which answer to choose" from a single-point judgment into a votable, aggregatable process, and is also the organization-structure source of the "weighted ensemble" in the quality curve.

The discuss–nominate–vote–select chain is worth taking apart because it corresponds to the three key links of ensemble learning. The **discussion** stage lets each expert see each other's intermediate conclusions, which essentially introduces inter-member correlation — if discussion makes everyone converge, then the diversity of subsequent voting falls, which is precisely the organizational source of homogenization. The **nomination** stage converges candidates to a few highest-scoring answers, equivalent to compressing voting from "all-member free divergence" to "a few representatives competing." The **voting** stage, by `hierarchical_select` taking the highest score within a team and then

across teams at the department level, is a tournament-style hierarchical aggregation. The **selection** stage fixes the final answer. The reason this paper's quality curve emphasizes "calibrated weighting rather than equal weight" is precisely that the "voting weight" in this chain should not be one person one vote but reflect each expert's historical accuracy on an independent calibration set — otherwise an always-wrong expert and an always-right expert have equal voice, producing the equal-weight drag-down phenomenon (0.2131).

4.3 The quality curve: the real weighting protocol (`quality_curve`)

Key honest point: the production path is **not** naive equal-weight averaging. `quality_curve` replicates the coordinator's real protocol:

```

modes = ("blind", "analytic")          # D=2 truly independent sources
calib_mse = each expert's MSE on an independent calibration set  # never peek at test
w = softmax(-calib_mse / tau), tau=0.05  # Eq.(2)
best_i = argmax(w)                     # strongest expert
for k in k_list:
    preds, weights = [], []
    for i in range(k):
        if i < D: add truly independent source i, weight w[i]
        else:      # homogeneous expansion: copy the strongest expert + zero-mean near-correlated noise
                    preds.append(test_pred[best_i] + N(0, sigma*0.01))
                    weights.append(w[best_i])
    mse(k) = mean( (Σ weights[i]*preds[i] - truth)^2 )

```

Where σ is 1% of the strongest expert's prediction standard deviation, i.e. the copy is near-correlated with the original expert ($\text{corr} \approx 0.9999$). This construction **deliberately simulates homogeneous expansion** — the added agent is almost the same person as the original expert. Three comparisons are also reported: `best_single` (best single expert), `naive_equal_2_mse` (equal-weight averaging of two sources, including the weaker source), `weighted_ensemble_mse` (calibrated weighting).

The weighting weights are computed by the following formula (softmax sharpness $\tau=0.05$):

$$w_i = \frac{\exp(-\text{calib_mse}_i / \tau)}{\sum_j \exp(-\text{calib_mse}_j / \tau)}, \tau = 0.05.$$

4.3.1 Why ensemble error decays as $a+b/k$ (intuitive derivation)

Consider the weighted-average prediction of k members $\hat{f}_k = \sum_{i=1}^k w_i f_i$. If member errors are pairwise independent, zero-mean, with variance σ^2 , then after averaging the variance falls to σ^2/k ; adding an irreducible bias term (system error shared by all members, not removable by averaging), the total MSE takes the form $a + \sigma^2/k = a + b/k$, i.e.

$$mse(k) = a + \frac{b}{k}, \quad \text{where } a \text{ is the irreducible floor, } b \text{ is the variance decline bought by independence.}$$

This is precisely the source of the $1/k$ law: a is the irreducible floor (determined by shared bias/strongest member), and b/k is the variance decline bought by error independence. The key corollary is: **when member errors are no longer independent (homogeneous), the variance term no longer falls with k , and b actually tends to 0** — at this point no matter how large k , MSE is pinned at a . This paper's homogeneous-copy construction ($\text{corr} \approx 0.9999$) precisely exposes this corollary: after

$k > 2$ the curve flattens, the measured embodiment of $b \approx 0$. This derivation explains the phenomenon in Table P9-1 where $a = 0.00925$ almost equals `best_single = 0.009196`: when all members come from the same near-correlated source, the room for ensemble variance decline is locked by correlation, and what remains is the strongest member's level.

Management implications of a and b . Translating fitted parameters into management language: a is the "irreducible floor" — it equals the level of the strongest person in the legion, and hiring more cannot get below a , because the remaining error is system bias shared by all members. b is the "diversity-dividend slope" — it depends on member-error independence; the larger b , the more different the people you hire are from each other, and the more error decline the ensemble can extract from diversity. This paper measures $b = -4.23 \times 10^{-5}$, very small in absolute value, precisely because after $k > 2$ added members are near-correlated with the original expert — the diversity dividend is wiped out by homogenization. This gives decision-makers a direct insight: rather than seeing b 's absolute value as large, first see whether a can be lowered (switch to a stronger strongest member), then see whether b can be enlarged (introduce truly independent information sources), rather than continuing to add homogeneous members after b has tended to zero.

4.4 Saturation fitting (`fit_quality_saturation`)

Do least-squares fitting of $mse = a + b/k$ over $(k, mse(k))$: the design matrix is $[1, 1/k]$, solved by `np.linalg.lstsq`. This paper measures $a = 0.0092520456$, $b = -4.2334 \times 10^{-5}$. This is a typical Kaplan-style practice: drawing the (k, y) scatter and fitted line together and concentrating fitted parameters into a lookup table (last row of Table P9-1), rather than giving only a curve.

4.5 Throughput curves (`throughput_curve` / `io_bound_curve`)

`throughput_curve` uses 48 independent prediction tasks, changes the coordinator concurrency `n`, and measures wall-clock and tasks/s. The CPU-intensive path runs real world-model forward passes. `io_bound_curve` uses `asyncio.sleep(latency=0.05)` to simulate remote LLM/tool waiting (not occupying CPU), measuring I/O-intensive scaling. The CPU-intensive path obeys Amdahl-type constraints:

$\text{speedup}(n) \setminus \text{peaks near physical parallelism} \setminus n_{\text{phy}} \setminus \text{and falls for } n > n_{\text{phy}} \setminus \text{due to contention/synchronization overhead}$. $\tag{3}$

4.6 Automation levels (`automation.py`: AL0–AL5)

The `AL` enum runs from AL0 (purely manual) to AL5 (recursive self-improvement loop). `CapabilityGate.require(level)` **explicitly raises `GateError`** when backends (LLM key, cloud context, RSI) are missing, rather than using local deterministic flows to impersonate AL4/AL5. This paper honestly marks accordingly: within the CPU sandbox only deterministic measurable implementations of AL1–AL3 are provided.

5 Experimental Setup

Per the v2 specification, the empirical chapter unfolds as "experimental setup → comparison → ablation/sensitivity → validity threats" (validity threats in the separate Chapter 8).

- **Task:** 48 independent trajectory-prediction tasks (world model `checkpoints7/worldmodel_v7.0.3.pt`); quality samples 48.
- **Hardware fingerprint:** Linux CPU; Python 3.12.11, torch 2.14.0+cpu, **CPU threads 2**, `device=cpu`; baseline total wall-clock 5.865 s.

- **Random seeds:** test_seed=2026, calib_seed=314, copy-noise rng seed=0; deterministic.
- **Evidence grade:** throughput curves are CPU-measured; quality curve $k \leq 2$ is verified (truly independent sources), $k > 2$ is cpu-proto (homogeneous copies); I/O curve is simulation (asyncio.sleep, not real LLM latency).
- **Comparison-arm design:** three-arm comparison — (i) best_single (best single-expert lower bound); (ii) naive_equal_2_mse (equal weight including the weak source, comparing "no weighting"); (iii) calibrated weighting $\text{mse}(k)$ (production path). The homogeneous anti-arm is the near-correlated copies for $k > 2$.
- **Reproduction:** `python scripts7/agent_scaling_bench.py`.

Why calibration-set isolation is key. The weighting weights come from each expert's MSE on an **independent calibration set** (calib_seed=314), while the test set (test_seed=2026) is used only for final evaluation. If the weights peeked at the test set, then "the weighted ensemble is better" would become a conclusion polluted by data leakage — you would in effect score your experts using the test answers. This paper strictly separates the calibration set from the test set: `_batch_data(max(16, n_samples//2), calib_seed)` separately constructs calibration data with another seed, and the weights `w=softmax(-calib_mse/tau)` do not touch test at all. This isolation is the premise for treating the "weighted ensemble" as a credible report rather than a data-leakage product. `tau=0.05` controls softmax sharpness: the smaller tau, the more weights concentrate on the strongest expert; the larger tau, the closer to equal weight. This paper fixes tau=0.05 and does no tuning-style fitting.

6 Results

6.1 The quality saturation law (H1, H2)

Table P9-1 Quality curve and saturation fit (source `reports7/agent_scaling.json` `quality_curve` / `quality_saturation_fit`; evidence grades marked row by row)

k	Weighted-ensemble MSE	Best single expert	Equal-weight two experts	Truly independent sources	Evidence
1	0.009196	0.009196	0.213141	1	verified
2	0.009196	0.009196	0.213141	2	verified
3	0.009357	0.009196	0.213141	2	cpu-proto
5	0.009220	0.009196	0.213141	2	cpu-proto
9	0.009227	0.009196	0.213141	2	cpu-proto
17	0.009223	0.009196	0.213141	2	cpu-proto
Fit	$a=0.009252$, $b=-4.23 \times 10^{-5}$ (Eq.(1), 6-point least squares)	best_single=0.009196	equal_2=0.213141	D=2	cpu-proto

There are three key points in reading the table. First, **the saturation floor $\$a \approx 0.00925\$$ almost equals the best single expert 0.009196**: this shows the ensemble's lower bound is determined by "the best member," and piling on more homogeneous members cannot break through the strongest single expert's level. Second, **after $\$k > 2\$$ the curve is basically flat** ($0.009357 \rightarrow 0.009220 \rightarrow 0.009227 \rightarrow 0.009223$), fluctuating slightly around $\$a\$$ — the added 15 homogeneous copies did not significantly pull MSE down from 0.0092, and H2 holds. Third, **the equal-weight two-expert MSE is as high as 0.2131**, about $23\times$ worse than the weighted ensemble: if experts are not weighted by calibration quality but naively averaged, one weak source seriously drags down the result.

Figure P9-1: Quality saturation law

Figure P9-1 The quality saturation law. The horizontal axis is ensemble agent count $\$k\$$ (log scale, six points $1/2/3/5/9/17$), and the vertical axis is the weighted ensemble's MSE on the test set (lower is better); solid scatter points are `quality_curve` measurements ($\$k \leq 2\$$ verified, $\$k > 2\$$ cpu-proto), the red dashed line is the least-squares fit $\$mse(k) = a + b/k\$$ (Eq.(1), $\$a = 0.009252\$$, $\$b = -4.23 \times 10^{-5}\$$), and the green dotted line is the best single-expert level 0.009196. Since $\$b < 0\$$, the fitted line slowly rises from about 0.009210 at $\$k = 1\$$ and asymptotes to $\$a = 0.009252\$$ (about 0.009250 at $\$k = 17\$$), rather than descending to the floor from above — the line is nearly horizontal, precisely depicting that homogeneous copies bring no variance decline. The equal-weight two-expert 0.2131 is far above the saturation region and is marked in text in the figure (not drawn at the same scale). Data source `reports7/agent_scaling.json`, single-seed pilot (test_seed=2026).

It must be honestly stated what direction and shape the fit has: since $\$b = -4.23 \times 10^{-5} < 0\$$, the fitted line numerically **slowly rises** from about 0.009210 at $\$k = 1\$$ and asymptotes to $\$a = 0.009252\$$ as $\$k\$$ grows (about 0.009250 at $\$k = 17\$$), rather than descending to the floor from above — this line is nearly horizontal ($\$|b|\$$ extremely small, $\$b \approx 0\$$), precisely depicting that homogeneous copies bring no variance decline. Each measured point jitters slightly within ± 0.00015 around $\$a\$$ (0.009357 at $\$k = 3\$$ is a noise peak), with no point significantly breaking below $\$a\$$.

Form comparison with the Kaplan power law (template integration). Kaplan's loss versus scale is a **power law** $\$L(N) = (N_c/N)^\alpha\$$, approximately continuously and monotonically falling across > 6 orders of magnitude with no upper turning point; this paper's $\$mse(k) = a + b/k\$$ is a **hyperbolic $\$1/k\$$ saturation**: it flattens to the irreducible floor $\$a\$$ at very small $\$k\$$ and is nearly horizontal due to homogenization with $\$b < 0\$$. The engineering implications of the two are opposite: the power-law narrative is "scale can still buy decline, just marginally slower"; this paper's saturation narrative is **"after the truly independent source count $\$D\$$, buying scale buys zero."** Kaplan in its 1.1 uses 8 bullets to give conclusions first and then dedicates Caveats in Section C; this paper follows this: the conclusion map is in § 1 and Caveats in the separate Chapter 8. We **do not carry any Kaplan exponent value (such as α_N) into this paper** — that is its own fitted constant, invalid across domains.

The mechanism of equal-weight failure. The figure `naive_equal_2_mse=0.213141` deserves separate explanation because it is the most counter-intuitive. It comes from doing an **unweighted arithmetic average** of the two predictions of the `blind` learned model and the `analytic` analytical baseline. When one source has a larger systematic bias, equal-weight averaging pours its bias into the final prediction at 50% weight, while calibrated weighting greatly lowers its weight according to calibration-set MSE. In other words, "one person one vote" is harmful when member quality is uneven — which is precisely the easiest pitfall in multi-agent voting scenarios: giving each agent equal voice equals making the weak agent as important as the good agent. This paper's measurement quantifies how large this pit is (0.2131 vs 0.0092, about $23\times$).

6.2 Throughput curves (H3, H4) — including the "throughput peak" negative result

Table P9-2 Throughput scaling (48 independent tasks, source `throughput_curve_cpu_bound / io_curve_remote_agent_simulation`; CPU threads=2)

Concurrent agents	CPU-bound wall_s	CPU tasks/s	CPU speedup	I/O-bound wall_s	I/O tasks/s	I/O speedup
1	0.3133	153.208	1.000	2.4071	19.94	1.000
2	0.1966	244.172	1.594	1.2038	39.88	2.000
4	0.3031	158.386	1.034	0.6023	79.69	3.997
8	0.4508	106.485	0.695	0.3014	159.26	7.986

Figure P9-2: Throughput curves

Figure P9-2 Throughput-parallelism dual curves. The horizontal axis is concurrent agent count n (1/2/4/8), the left vertical axis is throughput tasks/s (the right vertical axis is speedup relative to one agent). The solid line is CPU-bound local inference (constrained by this machine's 2 threads, peak 244.172 tasks/s @ $n=2$, speedup 1.594, after which $n=4/8$ falls to 158.386/106.485, speedup 1.034/0.695); the dashed line is I/O-waiting (simulated by `asyncio.sleep(0.05)`, simulation-grade evidence), near-ideally linear to 159.26 tasks/s, speedup 7.986. The red vertical line marks this machine's physical parallelism $n_{\text{phy}}=2$, i.e. where the CPU curve peaks. Data source `reports7/agent_scaling.json`, CPU measured / I/O simulated.

This is this paper's most important negative result. CPU-intensive path: **the peak is exactly at 2 agents** (244.172 tasks/s, speedup 1.594 $\times$), and adding to 4 agents makes throughput not rise but fall to 158.386 tasks/s (speedup only 1.034), with 8 agents further falling to 106.485 (speedup 0.695, i.e. slower than one agent). The reason is clear: this machine has only 2 CPU threads, the world-model forward is CPU-intensive, and beyond physical parallelism coordination overhead, the GIL, and torch thread contention instead become bottlenecks. This directly refutes the "more concurrency is faster" intuition.

Comparison with the I/O-intensive path: the same coordinator, just replacing each task with `asyncio.sleep(0.05)` to simulate remote waiting, throughput rises near-linearly from 19.94 to 159.26 tasks/s, with speedup reaching 7.986 $\times$ (close to the ideal 8 $\times$). This shows: **"whether expansion is possible" depends on task form** — CPU-intensive expansion peaks near the core count, and I/O-waiting expansion can scale near-linearly to the concurrency-quota upper limit.

Why 8 agents are slower than 1 agent. Looking row by row at the CPU column of Table P9-2: 1 agent 0.3133 s, 2 agents 0.1966 s, 4 agents 0.3031 s, 8 agents 0.4508 s. Wall-clock reaches its shortest at 2 agents and then monotonically lengthens. This is not measurement noise but has a clear mechanism: this machine has only 2 CPU threads and torch inference is CPU-intensive. When concurrency goes from 2 to 4, 8, multiple worker threads compete on the same set of physical cores, and adding the asyncio coordinator's scheduling and synchronization overhead, the effective parallelism of a single task does not rise but falls, with context switching and thread contention starting to dominate. This is precisely the intuitive embodiment of the Amdahl effect on a small machine: when the parallelizable part is constrained by physical core count, exceeding this point by adding concurrency only increases scheduling cost. It must be emphasized that we did not "decorate" this segment into a smoothly rising curve — the 8-agent 106.5 tasks/s is indeed lower than the 1-agent 153.2, and speedup 0.695 is an honest negative result.

6.3 Sensitivity and ablation: equal weight vs weighted, CPU vs I/O

Per the v2 specification, two comparison/sensitivity groups are reported centrally here:

- **Comparison A (weighted vs equal weight):** on the same pair of experts (`blind` learned model + `analytic` analytical baseline), calibrated weighting (Eq.(2)) gives 0.009196, equal-weight arithmetic averaging gives 0.213141, about $23\times$ different. This proves "whether expansion is useful" highly depends on "whether weighting by calibration quality" — without weighting, expansion (in fact merging a weak source) drags down the result.
- **Comparison B (CPU-intensive vs I/O-waiting):** the same coordinator, the same concurrency axis (1/2/4/8), the CPU curve peaks then falls (hump), and the I/O curve is near-ideally linear ($7.986\times$). The difference between the two is not in the coordinator but in whether the task form occupies physical cores.
- **Sensitivity (\$k\$ point positions):** the fitted a, b are based on 6 k values (1/2/3/5/9/17). Since the MSE for $k=3,5,9,17$ is almost flat around 0.0092, the fit is insensitive to small jitters at these points; but the first 2 points are verified and the last 4 are cpu-proto copy points, and the mixed evidence grade means the confidence intervals of a, b are not quantified. This sensitivity problem is explicitly retained as a validity threat in Chapter 8.

6.4 Automation-level comparison

Table P9-3 AL0–AL5 capability levels (source `automation.py: baseline_assessment`; evidence grades marked row by row)

Capability	Level	Evidence	Note
Deterministic trajectory prediction, single task	AL1	verified	Human initiates, model directly gives prediction
Multi-agent prediction ensemble/discussion/vote selection	AL2	verified	Agents assist, human defines goals and candidates
Coordinator high-level goal → decomposition → parallel verification → selection (narrow domain)	AL3	verified	End-to-end within goal templates, anomalies still need human review
Constrained code/config improvement (auto scoring within candidate set)	AL2	cpu-proto	Candidate strategy set is closed, not free programming
Free-form cloud thousands of LLM sub-agents writing code	AL4	unverified	Needs LLM key+cloud context, GateError if unconfigured
Recursive self-improvement (RSI) loop	AL5	unverified	Not provided by this repo, nor claimed

6.5 From two curves to the engineering criterion of "when to expand"

Putting the quality curve and throughput curve together, an operational decision tree can be extracted for engineering teams to judge "where this budget should go":

1. **First ask the task form.** If each agent's work is CPU-intensive (local forward, local computation), expansion is constrained by physical core count — throughput peaks and falls near the core count, at which point adding people does not increase revenue. If it is I/O-waiting (remote LLM calls, tool waiting), expansion is constrained by concurrency quotas and can scale near-linearly to the quota upper limit.
2. **Then ask the quality margin.** If added agents are near-correlated copies of existing members (same model, same prompt, same data), the quality curve flattens after the truly independent

source count, and expansion does not reduce MSE. At this point the budget should go to "introducing truly independent information sources" or "improving weighting," not copying existing members.

3. **Finally check weighting.** Naive equal weight lets weak experts drag down the result (measured 0.2131 vs 0.0092); softmax weighting must be done by quality on an independent calibration set, and the calibration set must not peek at the test set.

These three steps together are this paper's "failure boundary": **the quality boundary is determined by error independence, the throughput boundary by physical parallelism, and the two must be measured and decided separately.** A common error is to mistake "throughput can scale" for "quality can improve" — in I/O-intensive scenarios throughput can add people linearly, but if all added are homogeneous copies, quality is long pinned at the saturation floor $\$a\$$.

The `GateError` design is the core of the level gate. `CapabilityGate.require(level)` checks `Llm_keys_configured` and `cloud_context_configured` when requesting AL4, and additionally checks `rsi_loop_present` when requesting AL5; if any is missing it raises `GateError` and lists what is missing. This means the system does not "silently degrade" — it does not use local deterministic flows to impersonate "free-form cloud agents" without an LLM key, but explicitly refuses. This discipline of "explicitly error when the backend is missing, rather than pretending to do it" is the projection of this paper's whole evidence-grading system on the capability-autonomy dimension: AL levels are narrow-task scopes, not general autonomy declarations; each level marks the evidence grade (verified/cpu-proto/unverified) and the external switch needed to unlock it.

7 Discussion

7.1 The saturation floor is determined by the strongest member (not the average)

$\$a=0.00925\$$ almost equals `best_single=0.009196`, a conclusion with management implications: rather than spending the budget on "hiring 15 similar people," first find and use well that strongest expert. The marginal gain of homogeneous expansion is close to zero, even (if equal-weight) negative.

7.2 The "throughput peak" is bound to this machine's core count and cannot be extrapolated

The peak at 2 agents is because this machine has 2 threads; on a 16-core machine the CPU-intensive peak would appear at about 16 agents. Therefore this paper gives the "mechanism shape" (first rise then fall, peak near physical parallelism), not the absolute peak position. Across hardware it must be re-measured with the hardware fingerprint reported.

7.3 Homogeneous copies versus NSA's "independent pathways"

Borrowing the NSA template: its sparse attention preserves capability under a unified budget because the compression/selection/sliding-window three pathways each capture **independent** information; if two pathways attend to the same token, the second is waste. This paper's homogeneous copies are precisely the degraded case of "only one pathway, copied $\$k\$$ times" — $\text{corr} \approx 0.9999$ means added copies carry no new information. Therefore the architectural countermeasure is not "copy the same source more," but "as NSA gives three independent pathways, introduce truly independent information sources to the legion." This analogy does not carry NSA's speedup figures (which are `unverified` external figures for this paper).

7.4 Division of labor with P1 (topology)

This paper covers "saturation of quality and throughput versus k ," and P1 covers "decline of communication rounds versus N ." The two complement: P1 cares about "how to organize to reduce communication among a million agents from $O(N)$ to $O(\log N)$," and this paper cares about "in an already organized legion, whether adding one more agent is useful for quality and throughput." A legion may have a very efficient communication topology (P1's conclusion holds) but member homogenization causes quality to have long saturated (this paper's conclusion holds) — these two things do not contradict and correspond respectively to coordination overhead and member diversity, two different dimensions.

7.5 Why negative results should be kept

This paper deliberately keeps three "not pretty" results: CPU throughput at 8 agents is slower than one agent (speedup 0.695), homogeneous expansion does not reduce MSE, and equal-weight averaging is about $23\times$ worse than a single expert. If these results were "decorated" away, one would get a misleading chart of "adding people is always better." Our stance is: half the value of a scaling-law paper is drawing the rising segment, and the other half is honestly drawing the descending and plateau segments — the latter is the boundary engineering decisions truly need.

8 Limitations and Validity Threats (separate chapter, per Kaplan Section C Caveats)

This section, following the Kaplan template's dedicated Caveats, centrally reports the failure points this paper knows and "the most likely counterexamples + how this paper guards against them."

T1 Single-seed pilot threat (main). The quality curve is a fixed-seed point estimate (test_seed=2026, calib_seed=314), without CI; a, b are 6-point least-squares point estimates. **Most likely counterexample:** with another set of seeds, the fitted a, b will drift and the saturation-point position will move. **How this paper guards:** all conclusions are bounded as "single-seed pilot point estimates" and marked [RESULT NEEDED: multi-seed quality-curve CI]; the pre-registration plan requires ≥ 30 seeds per point (see § 9).

T2 The fitted model has no leave-one-out comparison. Currently only the one model $mse = a + b/k$ is fitted. **Most likely counterexample:** the power law $a + b/k^\beta$ with $\beta \neq 1$ may fit better. **How this paper guards:** H1 already writes "if β is significantly $\neq 1$ revise the claim" as the falsification condition; model-selection robustness is listed as to do (§ 9).

T3 The quality curve for $k > 2$ is cpu-proto, not a real heterogeneous legion. There are only 2 truly independent sources; the points $k = 3..17$ are near-correlated copies of "strongest expert + 1% standard-deviation noise," depicting homogeneous expansion and **do not represent** an ensemble of 17 real heterogeneous LLMs. **Most likely counterexample:** the declining segment of a real heterogeneous multi-vendor LLM ensemble may be steeper than this paper. **How this paper guards:** evidence grades marked row by row; a real heterogeneous legion needs multi-vendor API keys (the AL4 gate), listed as an unlock item.

T4 The throughput peak is bound to this machine's core count. The peak at 2 agents is due to this machine's 2 threads and cannot be extrapolated to 16 cores or GPU. Across hardware it must be re-measured with the hardware fingerprint reported.

T5 Narrow task spectrum. The tasks are this repo's trajectory prediction and do not represent open-domain reasoning. Extending conclusions to heterogeneous tasks such as code generation and research agents requires redoing the experiments.

T6 The I/O curve is simulation. `io_bound_curve` uses `asyncio.sleep(0.05)` to simulate remote waiting, measuring the scheduler's concurrency-scaling ability under waiting-type load, not the latency distribution of real LLM endpoints (real clouds have queuing, rate limiting, long tails, and network jitter). This paper only claims "the scheduling mechanism is consistent and the near-linear trend is reproducible," not "a real LLM legion can linearly accelerate to $7.986\times$."

T7 Fitting-quality self-assessment. Consistent with Kaplan's honest self-assessment that "the fits are imperfect": this paper's a, b fit is based on 6 points of mixed evidence grade (first 2 verified, last 4 cpu-proto), the extremely small $|b|$ makes the fitted line nearly horizontal, and the sign of b is sensitive to the copy-noise construction (the 1%-of- σ setting). This paper treats a, b as "parameterizations of mechanism shape," not "exact physical constants."

9 Resource Gates, Model-Selection Robustness, and Applicability Boundaries

Resource gates (currently reachable on CPU): organization tree (hundred-million closed-form staffing), seven-expert hierarchical voting, quality saturation curve, CPU/I-O throughput comparison, AL level gate.

Unlock items: the quality curve of a real heterogeneous LLM legion needs multi-vendor LLM API keys + cloud context (the AL4 gate); the RSI loop needs a verifiable self-improvement loop (AL5, not provided by this repo).

Model-selection robustness (to do). Currently only the one model $mse = a + b/k$ is fitted. Before submission three candidates should be compared: $a + b/k$ (this paper), power law $a + b/k^{\beta}$ (allowing the exponent to deviate from 1), logarithmic $a + b \log k$. Use leave-one-out to compare residuals over 6 points. The pre-registration rule is: if the 95% CI of the power-law β significantly excludes 1, revise the " $1/k$ " claim and report the real exponent; if the three models cannot be distinguished over 6 points (too few points), honestly report "model selection is unidentifiable, more k points needed," rather than hard-selecting one. This rule prevents us from reporting only it to make the " $1/k$ law" look good.

Pre-registration plan: ≥ 30 seeds per point; fitted models compared by leave-one-out among $a + b/k$, power law $a + b/k^{\beta}$, and logarithmic; the homogeneous arm as a negative control (expected no gain); no tuning to remove negative results.

Why the I/O curve is marked simulation. `io_bound_curve` uses `asyncio.sleep(0.05)` to simulate remote LLM/tool waiting; it measures the **scheduler's concurrency-scaling ability under waiting-type load**, not the latency distribution of real LLM endpoints. Real cloud LLM calls have queuing, rate limiting, long tails, and network jitter, and their speedup will deviate from the ideal line. Therefore this paper marks the I/O curve as `simulation`, claiming only "the scheduling mechanism is consistent with the coordinator and the near-linear trend is reproducible," not "a real LLM legion can linearly accelerate to $7.986\times$." This distinction is consistent with the declaration in the `notes` field of `agent_scaling.json`.

10 Conclusion

The expansion of an agent legion is not linear. On quality, ensemble MSE saturates as $\$a+b/k\$$ ($\$a=0.00925\$$, about equal to the best single expert), homogeneous expansion is ineffective after the truly independent source count, and equal-weight weak experts even drag down (0.2131 vs 0.0092); on throughput, CPU-intensive peaks at 2 agents (244.17 tasks/s, $1.594\times$) then falls, and I/O-waiting scales near-linearly to $7.986\times$. The conclusion is an engineering criterion: **first find the strongest expert and weight by calibration quality, then decide whether to expand by task form — CPU-intensive should stop near the core count, and only I/O type can add people near-linearly.** This paper honestly keeps two negative results and uses AL levels to brake the legion narrative.

Final response to the "headcount dividend" narrative. Multi-agent legions are not without value — they indeed reduce error when introducing truly independent information sources (which is precisely the source of the $\$b/k\$$ term), and under I/O-waiting load they indeed near-linearly raise throughput. But their value has strict boundaries: within the boundary, expansion is a dividend; outside it, expansion is waste or even negative return. Drawing this boundary clearly is more useful for engineering decisions than yet another curve chart of "adding people gets better." This is also why this paper puts "failure boundary" in the title.

One-sentence conclusion: the capability expansion of an agent legion obeys two independent saturation laws — on quality $\$mse=a+b/k\$$, whose floor $\$a\$$ is determined by the strongest member and slope $\$b\$$ by member independence, with homogeneous expansion's dividend going to zero after the truly independent source count; on throughput, CPU-intensive peaks and falls near physical parallelism, and only I/O-waiting can scale near-linearly. Accordingly, the correct expansion strategy is "first find the strongest member and weight by calibration quality, then decide the concurrency upper limit by whether the task is CPU-intensive or I/O-waiting," rather than indiscriminately piling on headcount. All figures in this paper are CPU fixed-seed pilot point estimates, with the I/O curve marked simulation and the $\$k>2\$$ quality points marked cpu-proto, not impersonating mechanism shape as exact physical constants or real heterogeneous-legion measurement — this restraint is where this paper differs from the "headcount dividend" promotional chart. For engineering decision-makers, what is truly operational is not "adding people," but "first find the strongest member and weight by calibration quality, then decide the concurrency upper limit by whether the task is CPU-intensive or I/O-waiting" — these two boundaries are worth more than any rising curve.

References

The first nine items below are  verified entries in this shared "Master Reference Library" (verification date 2026-09-19, fields copied from the master library); the rest are search directions to be supplemented, kept as placeholders, not fabricated. The bibliographic records of the three external templates (Kaplan / NSA / AlexNet) are based on the user's attached original PDFs, verification date 2026-09-20.

Verified (2026-09-19)

1. Kaplan, J., McCandlish, S., Henighan, T., et al. (2020). Scaling Laws for Neural Language Models. arXiv:2001.08361. (Related: loss falls approximately as a power law with $N/D/C$, the scale-scaling prototype of this paper's " $\$mse=a+b/k\$$ "; this paper only borrows its method, not exponent values.)

2. Hoffmann, J., Borgeaud, S., Mensch, A., et al. (2022). Training Compute-Optimal Large Language Models (Chinchilla). NeurIPS 2022. arXiv:2203.15556. (Related: about 400 model ablations, parameters and data scaling in proportion.)
3. Fedus, W., Zoph, B., Shazeer, N. (2021). Switch Transformers: Scaling to Trillion Parameter Models with Simple and Efficient Sparsity. JMLR 2021. arXiv:2101.03961. (Related: top-1 sparse MoE, about 1.6T parameters, about $7\times$ training acceleration.)
4. Krogh, A., Vedelsby, J. (1995). Neural Network Ensembles, Cross Validation, and Active Learning. NIPS 8 (NeurIPS 1995). (Related: bias-variance-ambiguity decomposition, the theoretical ancestor of this paper's "§b§ is determined by member independence.")
5. Wu, Q., Bansal, G., Zhang, J., et al. (2023). AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation. arXiv:2308.08155. (Related: conversable multi-agent framework, the comparison for this paper's agent legion.)
6. Amdahl, G. M. (1967). Validity of the Single Processor Approach to Achieving Large Scale Computing Capabilities. AFIPS SJCC 1967. (Related: the serial-fraction upper bound determines parallel speedup, the theoretical basis for this paper's CPU throughput peak.)
7. Vaswani, A., Shazeer, N., Parmar, N., et al. (2017). Attention Is All You Need. NeurIPS 2017. arXiv: 1706.03762. (Related: scaled dot-product attention makes "content-position-relation" a learnable connection, the classical starting point of single-model structural prior.)
8. Sennrich, R., Haddow, B., Birch, A. (2016). Neural Machine Translation of Rare Words with Subword Units (BPE). ACL 2016. arXiv:1508.07909. (Related: discretizing the open vocabulary into learnable subword units, input-side structural prior.)
9. Kudo, T., Richardson, J. (2018). SentencePiece: A simple and language independent subword tokenizer and detokenizer for Neural Text Processing. arXiv:1808.06226. (Related: pure-text-driven subword tokenizer decoupling tokenization from the model.)

External templates (user's attached original PDFs, verification date 2026-09-20; only methodological references, their values unverified for this paper)

- Yuan, Gao, Dai ... Ruan, Zhang, Liang, Zeng (DeepSeek-AI et al., 2025). Native Sparse Attention: Hardware-Aligned and Natively Trainable Sparse Attention. arXiv:2502.11089. (Related: the sparse-attention paradigm of compression/selection/sliding-window three independent pathways, the architectural comparison for this paper's "expansion should introduce independent sources rather than copies.")

To be verified (keeping [CITATION NEEDED] , not fabricated)

- Inference-time compute scaling: inference-time compute scaling test-time scaling.
- Multi-agent debate/voting orchestration: multi-agent LLM debate voting orchestration.

Appendix

Appendix A Reproduction commands and evidence ledger

```
python scripts7/agent_scaling_bench.py      # generates reports7/agent_scaling.json
pytest tests7/test_v72_legion.py -q        # 5 legion contract tests
```

Evidence ledger: main data reports7/agent_scaling.json (cpu-proto); source udos7/agents/{legion,automation}.py; hardware fingerprint python 3.12.11 / torch 2.14.0+cpu / 2 threads / ckpt worldmodel_v7.0.3.pt; baseline wall-clock 5.865 s.

Test index. tests7/test_v72_legion.py (5 items) covers legion contracts: organization-tree staffing computation, hierarchical selection, quality-curve monotonicity, CPU/I-O curve shape, and saturation-fit return structure. tests7/test_v71_agents.py (10 items) covers the seven-expert discuss/nominate/vote/select protocol and SharedMemory collision detection. These tests fix the "curve shape" rather than "some absolute figure" as regression assertions — e.g. the throughput curve should peak near physical parallelism rather than monotonically rise, and the quality curve should flatten after $k > D$ rather than persistently fall. Thus if later changes break the "saturation/peak" mechanism, the regression immediately alarms.

Appendix B Number traceability spot checks

1. CPU peak 244.172 tasks/s @ 2 agents $\leftarrow$ JSON headline.cpu_bound_peak.
2. Speedup 1.594 $\leftarrow$ JSON throughput_curve_cpu_bound[agents=2].speedup_vs_1.
3. Saturation $a=0.009252$ $\leftarrow$ JSON quality_saturation_fit.saturation_a.
4. Equal-weight two-expert MSE 0.213141 $\leftarrow$ JSON quality_curve[0].naive_equal_2_mse.
5. I/O speedup 7.986 @ 8 agents $\leftarrow$ JSON io_curve_remote_agent_simulation[agents=8].speedup_vs_1.
6. Fit $b=-4.2334 \times 10^{-5}$ $\leftarrow$ JSON quality_saturation_fit.gain_b.

Appendix C Author's intended-use statement

- **Target outlets:** NeurIPS/ICLR workshop, AAMAS, IEEE Software; journals TSE/EMSE (AI engineering perspective). Quartile/IF/deadline [to be verified], verified online before submission with the date marked.
- **Pre-registration plan:** ≥ 30 seeds per point, three fitted models compared by leave-one-out, homogeneous negative control.
- **Data and code availability:** Apache-2.0; tag v7.5.0 (commit 1a71270); report reports7/agent_scaling.json.
- **AI-use statement:** AI assisted in generating the draft and figure scripts, all figures sourced and checked by the author from JSON/source; unverified p-values/CI/DOIs/quartiles were not filled in; the I/O curve is explicitly marked simulation, and the $k > 2$ quality points marked cpu-proto.

Appendix D Internal review record (five-dimension reviewer self-assessment, look only, do not change)

1. **Novelty:** 3/5. The saturation law and Amdahl are both classical; the novelty lies in simultaneous calibration in the agent-legion scenario + homogeneous negative control.
2. **Evidence strength:** 3/5. CPU throughput is really measured (credible), but the quality curve for $k > 2$ is cpu-proto copies, single seed without CI.
3. **Reproducibility:** 5/5. Fixed seeds, hardware fingerprint, reproduction commands, complete evidence ledger.
4. **External validity:** 2/5. Narrow task spectrum, bound to this machine's core count, real heterogeneous LLM legion not run (stated).
5. **Writing and honesty:** 5/5. Keeps three negative results of "throughput peak then fall," "homogeneous ineffectiveness," and "equal-weight drag-down," AL levels braking the narrative.

Overall judgment: an honest empirical-measurement paper, suitable for workshops/experience reports; submission to a main conference needs multi-seed CI, power-law model comparison, and real heterogeneous-LLM legion experiments (unlocking Gate B).